Math 257 Definitions and Theorems

Week 12: Modules 39-40

Fall 2023

39 Gram-Schmidt Method

Theorem 39.1. *Every subspace of $\mathbb{R}^n$ has an orthonormal basis.*

Algorithm *(Gram-Schmidt orthonormalization)*: Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_m$, we produce an *orthogonal basis* $\mathbf{b}_1, \ldots, \mathbf{b}_m$ and an *orthonormal basis* $\mathbf{q}_1, \ldots, \mathbf{q}_m$.

$$\begin{array}{ll} \mathbf{b}_1 = \mathbf{a}_1, & \mathbf{q}_1 = \frac{\mathbf{b}_1}{\|\mathbf{b}_1\|} \\ \mathbf{b}_2 = \mathbf{a}_2 - \underbrace{\text{projSpan}(\mathbf{q_1})\mathbf{a}_2}_{=(\mathbf{a}_2\cdot\mathbf{q}_1)\mathbf{q}_1}, & \mathbf{q}_2 = \frac{\mathbf{b}_2}{\|\mathbf{b}_2\|} \\ \mathbf{b}_3 = \mathbf{a}_3 - \underbrace{\text{projSpan}(\mathbf{q}_1, \mathbf{q}_2)\mathbf{a}_3}_{(\mathbf{a}_3\cdot\mathbf{q}_1)\mathbf{q}_1 + (\mathbf{a}_3\cdot\mathbf{q}_2)\mathbf{q}_2} & \mathbf{q}_3 = \frac{\mathbf{b}_3}{\|\mathbf{b}_3\|} \\ \ldots & \ldots \end{array}$$

Remark. Notice that:

- $\text{Span}(\mathbf{q}_1, \ldots, \mathbf{q}_i) = \text{Span}(\mathbf{a}_1, \ldots, \mathbf{a}_i)$ for $i = 1, \ldots, m$, and
- $\mathbf{q}_j \notin \text{Span}(\mathbf{a}_1, \ldots, \mathbf{a}_i)$ for all $j > i$.

Theorem 39.2. (QR decomposition) *Let A be an $m \times n$ matrix of rank n. There is is an $m \times n$-matrix Q with orthonormal columns and an upper triangular $n \times n$ invertible matrix R such that $A = QR$.*

40 The Spectral Theorem

Theorem 40.1. *Let A be a symmetric $n \times n$ matrix, then A has an orthonormal basis of eigenvectors.*

Theorem 40.2. *Let A be a symmetric $n \times n$ matrix. Then there is a diagonal matrix D and a matrix Q with orthonormal columns such that $A = QDQ^T$.*